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ComplexityWorld: Benchmarking Vision-Language Models on Verifiable Visual Decision Making

390 Tasks Across 39 Visual Worlds Show That Most VLMs Cannot Extract Constraints from Images and Satisfy Them Jointly — Only One Model Exceeds 75%

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Many real-world AI tasks require more than perceiving what an image contains: a model must use visual evidence to make a decision whose components jointly satisfy global constraints. ComplexityWorld introduces a benchmark of 390 tasks across 39 domain-inspired visual worlds and 29 decision categories, each scored by an executable verifier that accepts any feasible solution. Under direct visual inference, all evaluated VLMs except GPT-5.6-Sol fall below 40% verifier acceptance rate (VAR), while GPT-5.6-Sol reaches 75.6%. A 23-point gap between visual and structured-text conditions reveals that the dominant failure mode is visual constraint extraction, not reasoning.

AT A GLANCE

Problem: VLM benchmarks test perception and QA accuracy but not the ability to make constraint-satisfying decisions from visual evidence, leaving a large deployment-relevant gap unmeasured.

Why it matters: Real-world tasks (scheduling, logistics, design) require visually-grounded constraint satisfaction; current evaluations do not capture whether models can do this.

Method: Generate 390 tasks from hidden structured specifications rendered as visual scenes; use executable verifiers rather than LLM judges to score feasibility.

Result: Only GPT-5.6-Sol exceeds 40% VAR (reaching 75.6%); all other models remain below 40%.

Evidence: Same tasks in structured-text form achieve +23 pp average VAR, isolating visual extraction as the key failure mode.

The Big Picture

Vision-language models have advanced rapidly on perception tasks — they can accurately describe image contents, count objects, and answer factual questions about visual

scenes. This progress has driven optimism about deploying VLMs in real-world decision support: assisting with medical imaging, factory floor monitoring, logistics planning, and scientific data analysis.

These real-world tasks share a property that standard benchmarks do not test: they require making a *decision* from visual evidence, where the decision must satisfy multi-part constraints. A scheduling assistant reviewing a visual calendar cannot merely describe meetings; it must identify conflicts, dependencies, and resource constraints from the visual layout and produce a feasible schedule satisfying all of them simultaneously.

This class of problem — *visual constraint-satisfying decision making* — is distinct from visual perception and from verbal reasoning, and it is largely unmeasured by current evaluation suites. ComplexityWorld is the first systematic benchmark for this capability.

Why Existing Approaches Fall Short

Perception benchmarks (VQA, object counting, scene description) test whether models correctly identify what is present in an image. They do not test whether models can extract structured decision-relevant information from visual representations of constraints.

Verbal reasoning benchmarks (GSM8K, logical reasoning datasets) test constraint satisfaction in text, but cannot assess whether visual extraction precedes and enables correct reasoning.

Prior VLM decision-making benchmarks have used LLM-as-judge evaluation, which is susceptible to hallucination and cannot distinguish a feasible from a plausible-sounding infeasible solution. ComplexityWorld’s verifier-based evaluation avoids these issues.

Core Mechanism

Task generation pipeline: Each task starts from a *hidden structured specification* $\mathcal{S}$ that defines:

- A set of objects, resources, or agents with attributes
- A set of hard constraints over decision variables
- A decision space (assignments, routes, orderings, partitions)

The specification is rendered into a visual scene I using a domain-appropriate renderer (grid-based for scheduling, graph-based for routing, CAD-based for spatial arrangement). An executable verifier $\mathcal{V}$ checks whether any response r satisfies all constraints in $\mathcal{S}$.

Key design choices: The verifier accepts *any* feasible solution, not just the canonical one, avoiding benchmark contamination and penalizing models solely for constraint violations rather than stylistic deviation.

Technical Formulation

Let $\mathcal{S} = (X, C, \mathcal{D})$ be a task specification with decision variables $X = \{x_1, \dots, x_n\}$, constraints $C = \{c_1, \dots, c_m\}$ over X , and domain $\mathcal{D}$.

A response r parsed from the model’s text output is a partial or complete assignment $\hat{X} = \{\hat{x}_1, \dots, \hat{x}_n\}$. Verifier acceptance:

$$\text{Accept}(r, \mathcal{S}) = \mathbf{1} \left[\bigwedge_{j=1}^m c_j(\hat{X}) = \text{TRUE} \right]$$

The verifier acceptance rate (VAR) for model M over task set $\mathcal{T}$:

$$\text{VAR}(M) = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \text{Accept}(M(I_i), \mathcal{S}_i)$$

The visual constraint extraction gap is measured by the difference between structured-text and visual-scene performance:

$$\Delta_{\text{VCE}}(M) = \text{VAR}(M, \text{structured}) - \text{VAR}(M, \text{visual})$$

Learning or Inference Procedure

ComplexityWorld is an evaluation benchmark, not a training procedure. Models are evaluated zero-shot with a standardized prompt that specifies the task domain, output format, and response constraints. No few-shot examples are provided in the main evaluation; ablations with $k = 1, 3, 5$ in-context examples show improvement but do not close the gap with structured-text performance.

What the Guarantee Says

The verifier is formally specified for each domain using executable constraint logic. Verifier correctness is validated by human experts who confirm that feasibility checks match domain semantics. This guarantees that VAR measures genuine constraint satisfaction, not LLM-judge agreement with plausible-sounding responses.

Experimental Findings

- **Direct visual inference VAR:** GPT-5.6-Sol reaches 75.6%; all other models remain below 40%.
- **Visual constraint extraction gap:** Average $\Delta_{\text{VCE}} = +23.1$ pp when the same task is given in structured text vs. as a visual scene.
- **Presentation sensitivity:** Changing visual style (color scheme, font, layout) while keeping constraints identical shifts VAR by up to 18 pp for the same model, revealing brittleness in visual feature extraction.

- **Category variation:** Routing and scheduling domains are hardest (mean VAR < 25%); assignment and partitioning domains are easier (mean VAR ~38%).
- **Open vs. proprietary models:** Best open-weight model achieves 31.4% VAR; best proprietary model (GPT-5.6-Sol) achieves 75.6%, a gap of 44.2 pp.

Ablations and Interpretation

In-context examples: Adding $k = 5$ in-context (image, decision) examples improves VAR by 6.2 pp on average, but narrows the visual-structured gap by only 1.3 pp, confirming that the bottleneck is visual extraction, not few-shot learning of the decision format.

Chain-of-thought prompting: Requiring models to reason step-by-step before producing a decision improves VAR by 4.8 pp on average, with larger gains on routing (+8.1 pp) and scheduling (+7.3 pp) tasks that have natural sequential decision structure.

Constraint complexity: VAR decreases sharply with the number of constraints m : from 52% at $m = 2$ to 21% at $m = 6$ for a representative model, suggesting exponential constraint combinatorics as the dominant difficulty driver.

Error analysis: Manual inspection of 200 failures reveals three failure modes: misreading a constraint value from the image (43%), correctly extracting constraints but producing an infeasible assignment (38%), and producing a feasible partial assignment that misses some variables (19%).

Reference

ComplexityWorld Authors. **ComplexityWorld: Benchmarking Vision-Language Models on Verifiable Visual Decision Making.** arXiv:2608.07584, August 2026. <https://arxiv.org/abs/2608.07584>